



GigaCloud AD Sync Agent

Configuration and Use Guide

July 2019

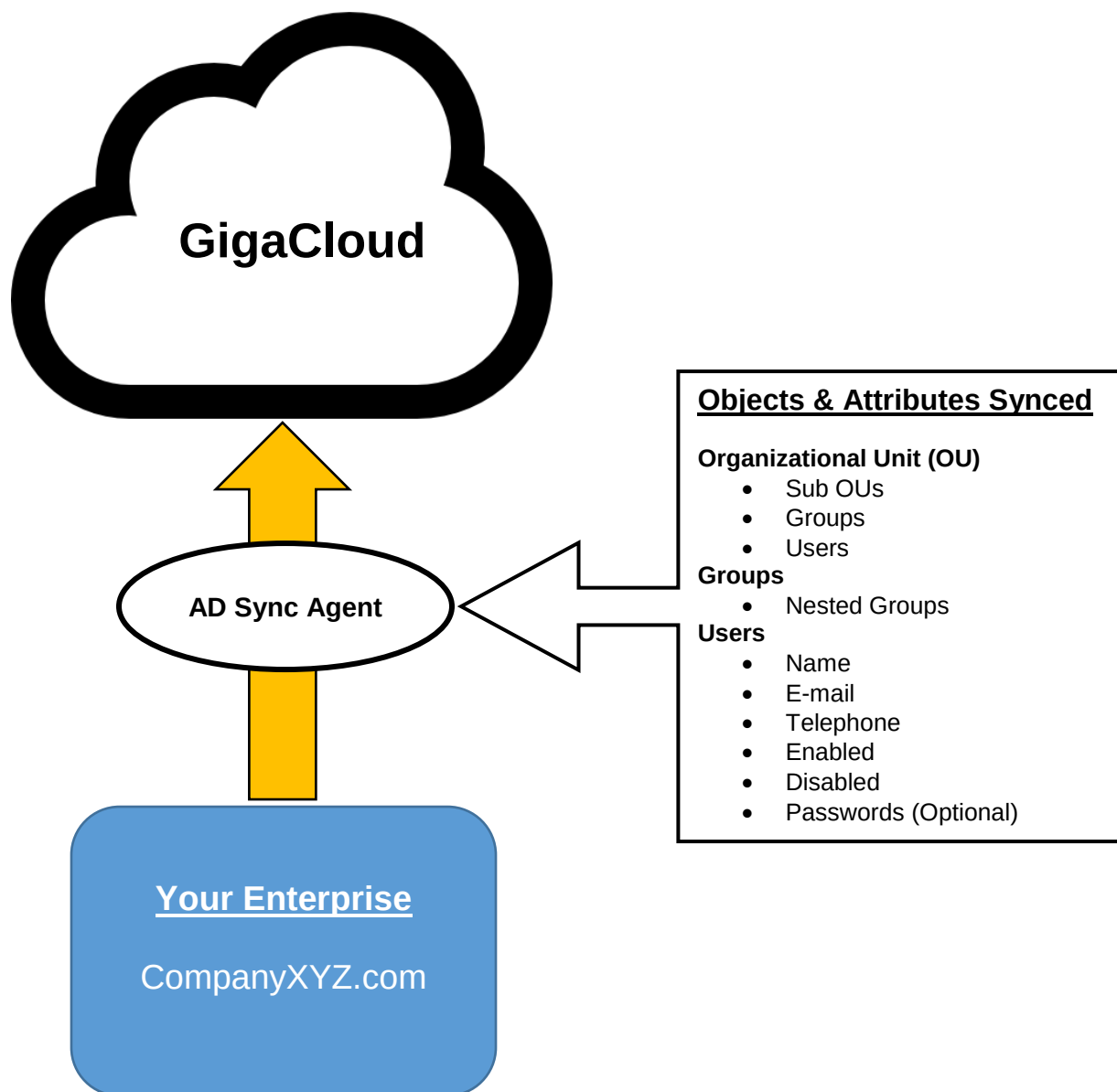
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Introduction

GigaCloud Active Directory Sync Agent (AD Sync Agent) is an autonomous tool that synchronizes domain forest Active Directory Users with GigaCloud's Active Directory services. As a Tenant of GigaCloud you can choose to synchronize User names and password hashes eliminating the need to manually add and manage them. Your local password management rules will be used in GigaCloud. It also allows Users to use the same password for your domain forest and GigaCloud.

The Sync Agent only syncs Users and Groups from your domain controller and copies them to your Tenant Active Directory container in GigaCloud. The actual Organizational Units (OUs) are not synced. Each object synced is copied to your Tenant folder root and group membership is maintained for all imported Users. If you delete a Group locally, on the next sync, that particular Group will be deleted in your Tenant folder as well as all User memberships to that group (but not the actual Users.) Maintaining Group memberships in GigaCloud allows you to apply RMS Policy Templates and Rules at a group level, thereby simplifying your AD RMS policy management.



GigaCloud AD Sync Console			
Connected: testcloud.trustedoffice.net			
Domain	Password	Deactivated Users	OU
syncme.local	☒	☐	OU=0723OU,DC=syncme,DC=local
syncme.local	☒	☐	OU=CTCGO1,DC=syncme,DC=local
syncme.local	☒	☐	OU=CTCGO2,DC=syncme,DC=local
syncme.local	☒	☐	OU=Chris TestOU,DC=syncme,DC=local
syncme.local	☒	☐	OU=Dan6,DC=syncme,DC=local
syncme.local	☒	☐	OU=Demo Accounts,DC=syncme,DC=local
Message			
Message		Timestamp	Exception
Completed password synchronization job for OU 'OU=0723OU,DC=syncme,DC=local'.		7/26/2019 2:46:09 PM	
Starting password synchronization job for OU 'OU=0723OU,DC=syncme,DC=local'.		7/26/2019 2:46:09 PM	
Completed password synchronization job for OU 'OU=darrenou,DC=syncme,DC=local'.		7/26/2019 2:45:44 PM	
Starting password synchronization job for OU 'OU=darrenou,DC=syncme,DC=local'.		7/26/2019 2:45:43 PM	
Completed password synchronization job for OU 'OU=TomOU,DC=syncme,DC=local'.		7/26/2019 2:45:43 PM	
Starting password synchronization job for OU 'OU=TomOU,DC=syncme,DC=local'.		7/26/2019 2:45:43 PM	
Completed password synchronization job for OU 'OU=OldDomainControl1,DC=syncme,DC=local'.		7/26/2019 2:45:43 PM	
Starting password synchronization job for OU 'OU=OldDomainControl1,DC=syncme,DC=local'.		7/26/2019 2:45:43 PM	
Completed password synchronization job for OU 'OU=Demo Accounts,DC=syncme,DC=local'.		7/26/2019 2:45:43 PM	
Starting password synchronization job for OU 'OU=Demo Accounts,DC=syncme,DC=local'.		7/26/2019 2:45:42 PM	
Completed password synchronization job for OU 'OU=Dan6,DC=syncme,DC=local'.		7/26/2019 2:44:17 PM	
Starting password synchronization job for OU 'OU=Dan6,DC=syncme,DC=local'.		7/26/2019 2:44:17 PM	
Completed password synchronization job for OU 'OU=CTCGO2,DC=syncme,DC=local'.		7/26/2019 2:44:17 PM	
Starting password synchronization job for OU 'OU=CTCGO2,DC=syncme,DC=local'.		7/26/2019 2:44:17 PM	

GigaCloud Active Directory Sync Agent v9.2 has the following features:

- 🔧 A simple to use console and tool bar menu.
- 🔧 Automated permission prompts eliminating the need to manually add required permissions to the service account.
- 🔧 Customizable sync job intervals.
- 🔧 Integrated GigaTrust Log Client allows real-time troubleshooting including Debug option for more detailed logging.
- 🔧 Optional Password synchronization keeps them hashed for maximum security.
- 🔧 Only sync Users that need to utilize your GigaCloud service account; leave other Active Directory objects locally secure.
- 🔧 Optionally sync deactivated Users.

Prerequisites

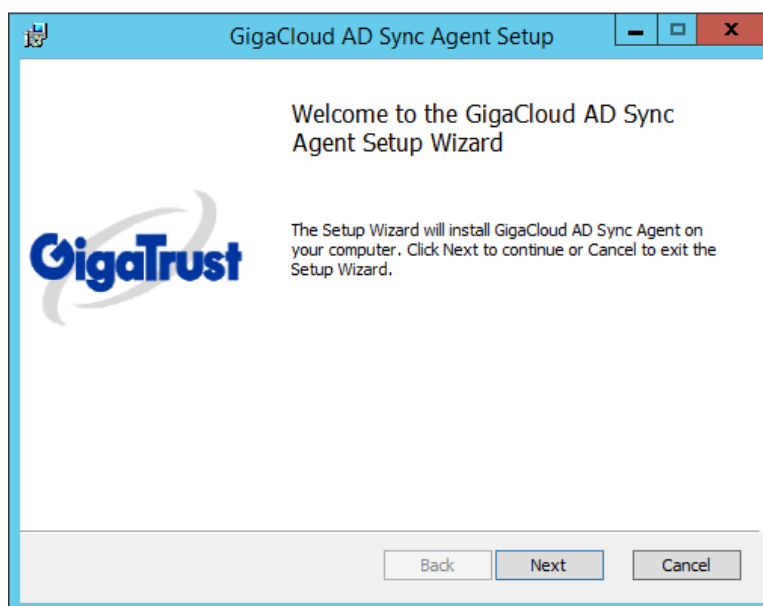
-  A server or desktop running Windows Server 2012 R2 / 2016 / 2019, Windows 8.1, Windows 10 (64 bit versions ONLY) joined to the forest to be synced.
-  Windows Server 2003/2008 domain controllers are supported if the Sync Agent is installed on a supported platform and is domain joined.
-  Microsoft .NET v4.6.2 Installed
-  An Active GigaCloud account that is a GigaCloud Tenant Admin. This user should NOT be in an AD OU that is Sync'd to GigaCloud.

Installation

The GigaCloud Active Directory Sync tool can be installed on any server in your forest, if it is joined to the domain. Ensure that Microsoft .NET v4.6.2 is installed and that there is enough RAM and CPU capacity. If you are installing on a Windows desktop, you should login, install and run the application as a Domain Administrator. You must also have an activated GigaCloud account.



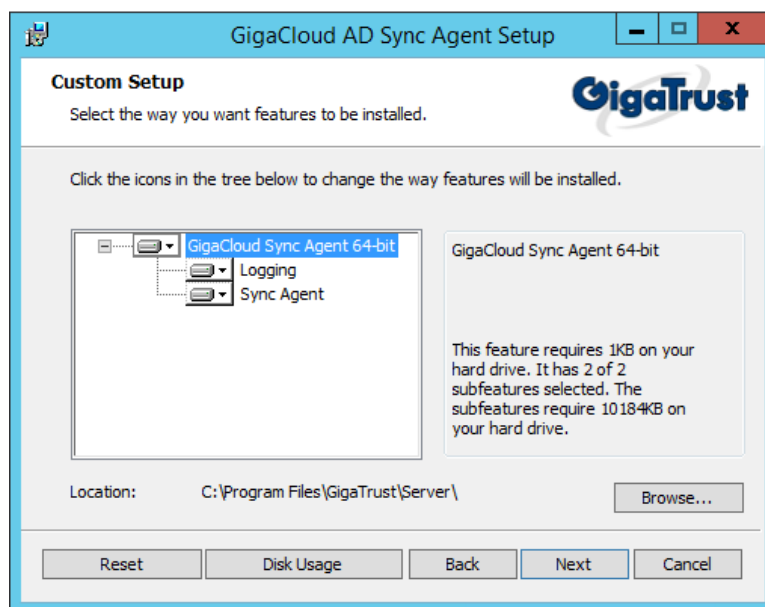
Important: It is advised that you create a new domain User account as the Sync Agent service account. The service account needs to be a local administrator and have local permissions to “logon as a service.” The service account should also have the password set to: “Password never expires” and should not be synchronized with your other synced users.



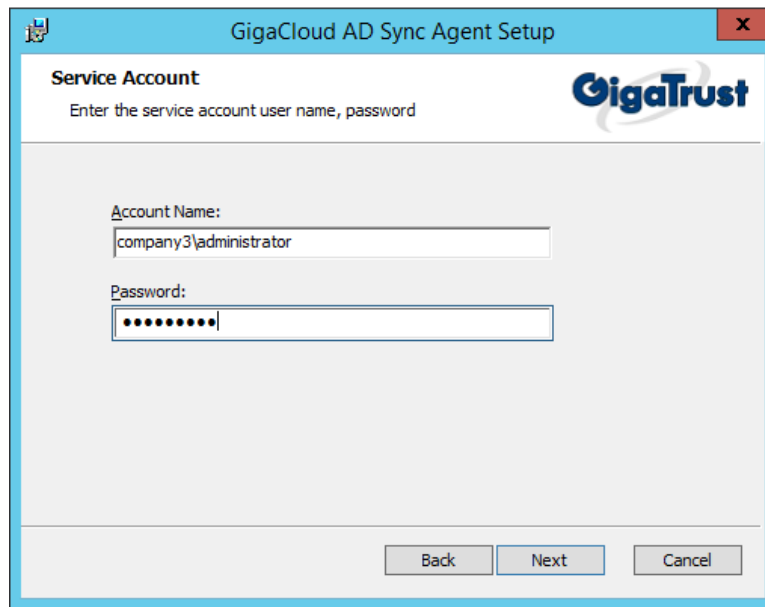
1. Click on the GigaCloud AD Sync Agent to start installation. Click **Next**.



2. Accept the End User Software License Agreement and select **Next**.

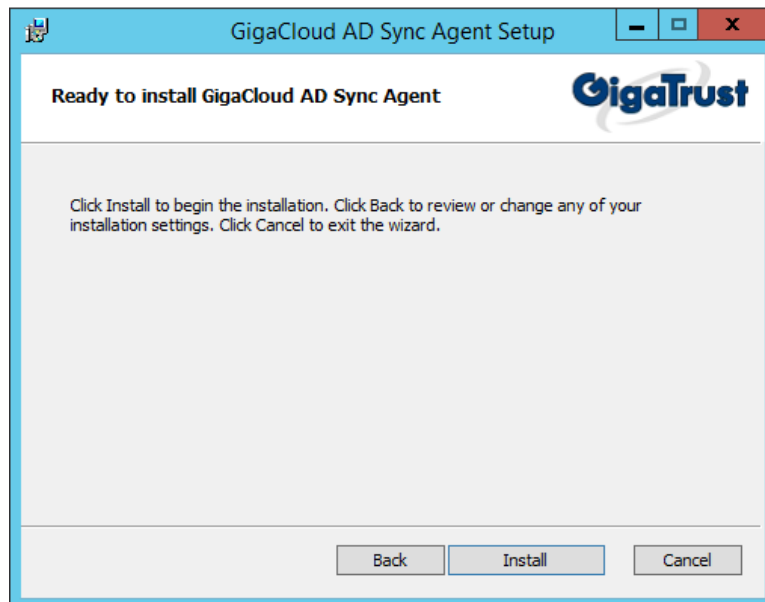


3. Leave both default features selected and click **Next**.



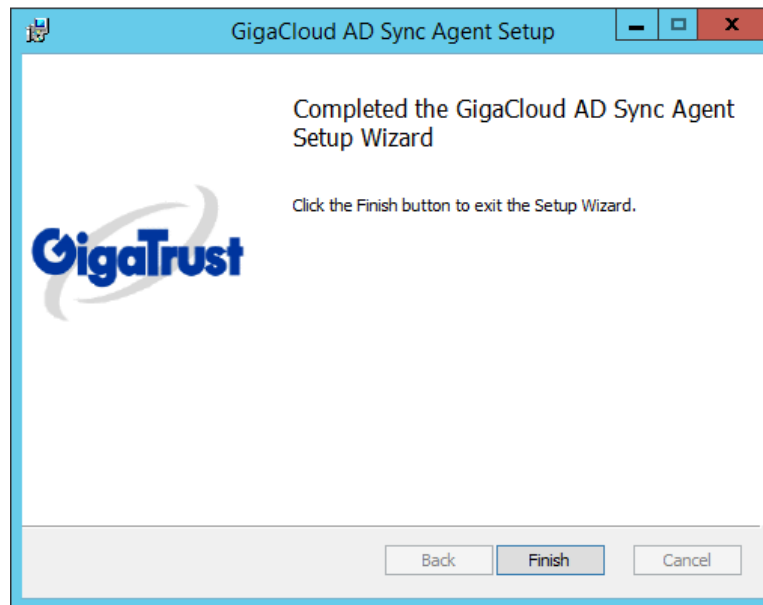
The screenshot shows the 'Service Account' step of the 'GigaCloud AD Sync Agent Setup' wizard. The window title is 'GigaCloud AD Sync Agent Setup'. The GigaTrust logo is in the top right. Below the title bar, the text 'Service Account' is followed by the instruction 'Enter the service account user name, password'. There are two input fields: 'Account Name:' with the text 'company3\administrator' and 'Password:' with masked characters. At the bottom, there are three buttons: 'Back', 'Next', and 'Cancel'.

4. Enter a domain user name and password that is also a local Administrator and has permissions to “logon as a service” and select **Next**.

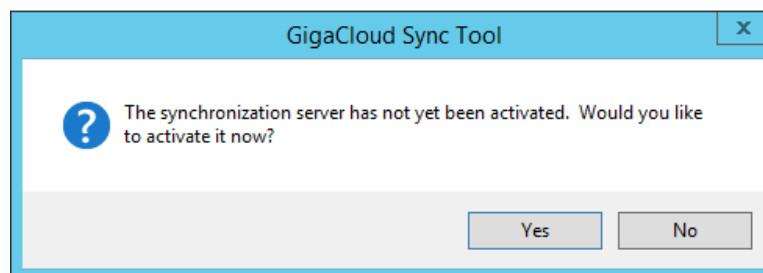


The screenshot shows the 'Ready to install GigaCloud AD Sync Agent' step of the 'GigaCloud AD Sync Agent Setup' wizard. The window title is 'GigaCloud AD Sync Agent Setup'. The GigaTrust logo is in the top right. Below the title bar, the text 'Ready to install GigaCloud AD Sync Agent' is followed by the instruction 'Click Install to begin the installation. Click Back to review or change any of your installation settings. Click Cancel to exit the wizard.' At the bottom, there are three buttons: 'Back', 'Install', and 'Cancel'.

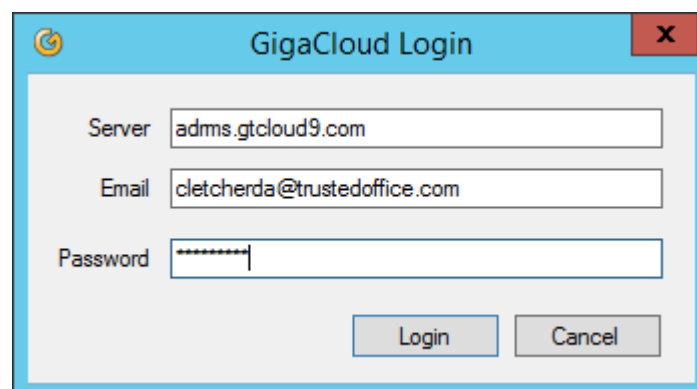
5. Click **Install**.



6. Click **Finish**.



7. The GigaCloud Sync Tool window prompts you to Activate the server for synchronization. Select **Yes** if you would like to configure it now. **No** if you plan on configuring it later.



8. If you selected **Yes** on the previous window, enter your GigaCloud information:
 - **Server** – Enter the GigaCloud address provided by your GigaTrust Sales Engineer. (i.e. adrms.gigacloud.com) Do not include "http(s)://".

- **Email** – Enter your GigaCloud Tenant Administrator email address.
- **Password** – Enter your GigaCloud password here.

Click on **Login**.

9. If login is successful the **GigaCloud AD Sync Console** window will be displayed. Continue configuring using the **Configuration** section below.



Note: If you are using an account that is not a Domain Administrator on a desktop client, the initial configuration window will not display, and the Console window will not start automatically. You will need to [configure permissions using the steps here](#) for the current User to enable the application to be started by this User.

Configuration

If you have already activated the server for synchronization, use the following steps to start synchronizing your GigaCloud Users from your local forest. It is recommended that you open the GigaTrust Log Client and connect to Localhost before syncing with your Tenant account in GigaCloud. Go to your programs list and choose the shortcut to open the Log Client. For more information view the [Troubleshooting](#) section at the end of this guide.



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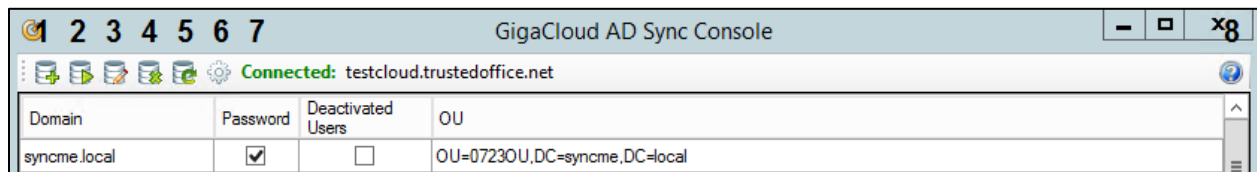
The console window is divided into two panels. The top panel will display synced OUs from your domain forest and include the following columns:

- **Domain** – the name of the forest/domain where the OU is synced from in your Enterprise.
- **Password** – indicates if User passwords are synchronized with GigaCloud.
- **Deactivated Users** – indicates if Deactivated Users are synchronized with GigaCloud.
- **OU**- the Active Directory location CN of the OU.

The bottom panel displays Sync job activity with the following columns:

- **Message**- displays relevant information for the job attempted or completed.
- **Timestamp**- UTC timestamp of the job that was run.

- **Exception-** any exception if the job failed. If there was a job error the row will be red.



The top tool bar provides the following options:

1. Click to add an OU to synchronize with GigaCloud.
2. This will queue up the job to synchronize the selected OU.
3. This allows you to configure settings for a selected OU.
4. Click to delete an OU from GigaCloud. Select the OU entry in the panel below before clicking to delete it. You will need to confirm deletion.
5. This will refresh the OU table.
6. You can configure AD Sync Agent settings here.
7. This displays whether the Sync Agent is connected and to which GigaCloud instance.
8. This will open an HTML help file.

More detailed instructions on the tool bar features can be found in the [OU Synchronizing](#) section.

OU Requirements



Important: Before you can synchronize Users in an OU, there is one User requirement: each User must have the **E-mail** attribute set as shown below displayed in the **General** tab in the User Properties window.

The screenshot shows the 'user2 Properties' window with the 'General' tab selected. The 'E-mail' field is highlighted with a red circle and contains the text 'user2@company3.local'. Other fields include First name (user2), Last name, Display name (user2), Description, Office, Telephone number, and Web page. The window has standard Windows controls at the bottom: OK, Cancel, Apply, and Help.



Note: Not all Group and User attributes are synchronized. View the [Appendix](#) for a detailed list of required and optional attributes that are synced. Additionally, Group names in GTEP will have the local Group name appended with your GigaCloud Tenant name – see [Group Naming](#) section for details.

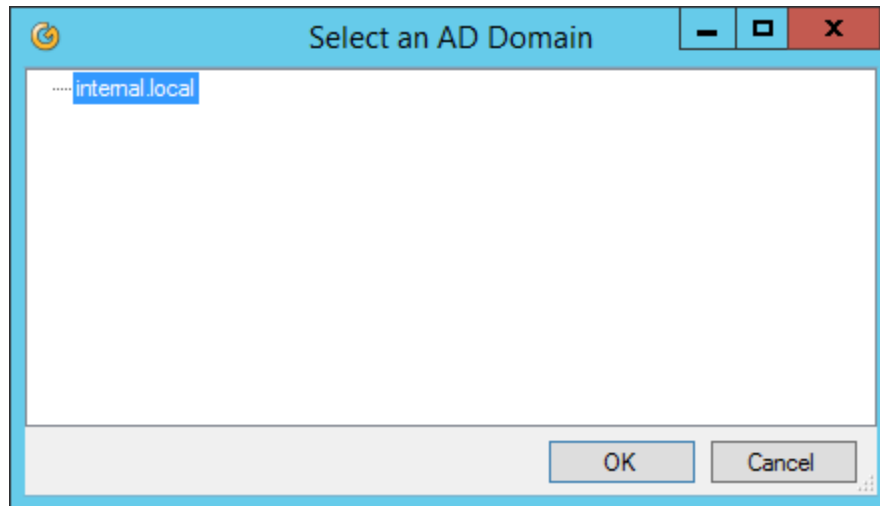


Important: Sync'd OU's will automatically consume a GigaCloud Client license for each user when they activate their GigaCloud account. You should only sync Users in OUs that require a GigaCloud account otherwise your license use count will be inaccurate.

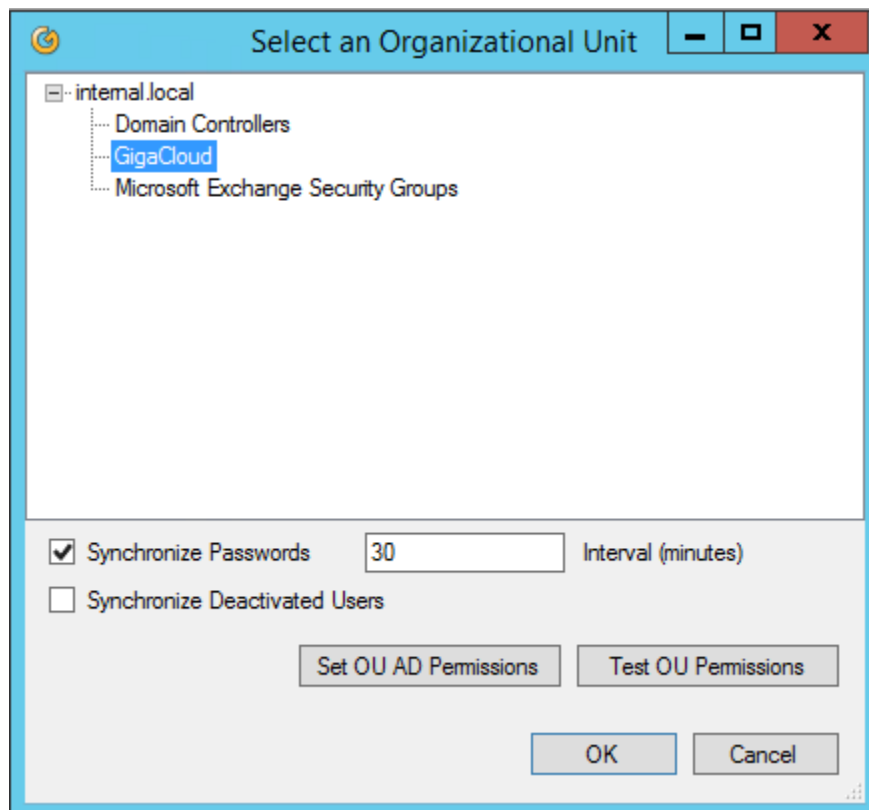
OU Synchronizing

To add an OU for Syncing do the following:

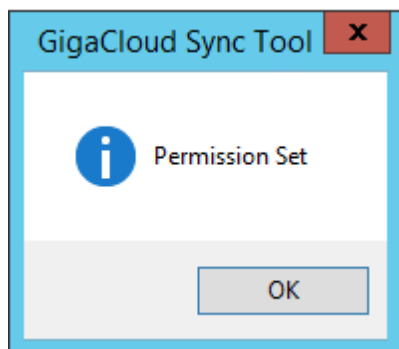
1. Click on the plus sign icon in the toolbar. 



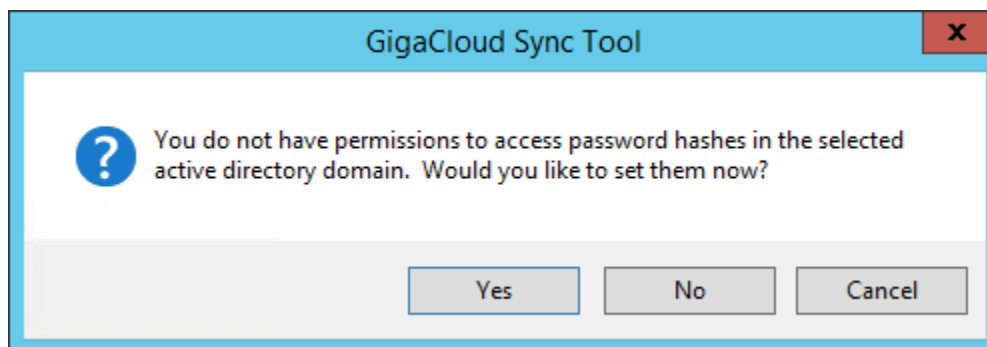
2. The **Select an AD Domain** window is displayed. All Domains (forests) will be displayed there. Choose the domain where the OU you need to add is located and select **OK**.



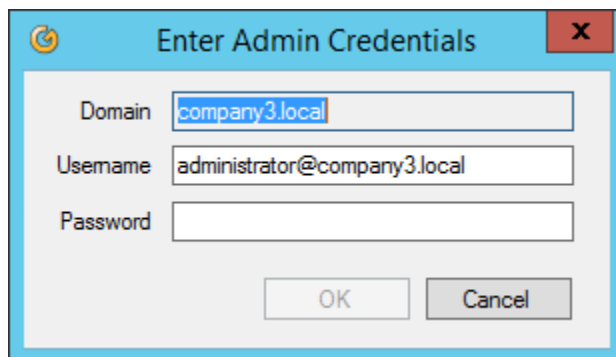
3. The **Select an Organizational Unit** window is displayed containing all of the OUs and Domain Controllers in your Active Directory.
- Choose the OU/Domain Controller you need to sync with GigaCloud. It is recommended you use an OU/OUs for syncing rather than the entire Domain Controller.
 - Synchronize Passwords-** this option is pre-selected for you. By synchronizing user passwords, the management of users is much less time consuming. Password expiration will still be handled by your local Active Directory and changes will be synced with GigaCloud dependant on the sync interval.
 - Interval (minutes)-** Customize the duration (or interval) in minutes for syncing between your Active Directory and GigaCloud. It is recommended for performance reasons to not set any less than 5 minutes, especially if you have a large number of Users (thousands of users).
 - Synchronize Deactivated Users-** This option is NOT pre-selected for you. Users Deactivated in your Active Directory will not sync to your tenancy unless you select this option. By not synchronizing deactivated users, the synchronization jobs will complete faster and will keep your tenant entries cleaner.
 - Set OU AD Permissions-** This button will set the necessary Active Directory permission to allow syncing of the selected OU and parent OUs.



- Test OU Permissions-** Test the permissions were applied to the selected OU to permit syncing with GigaCloud.
- Select **OK** when finished.



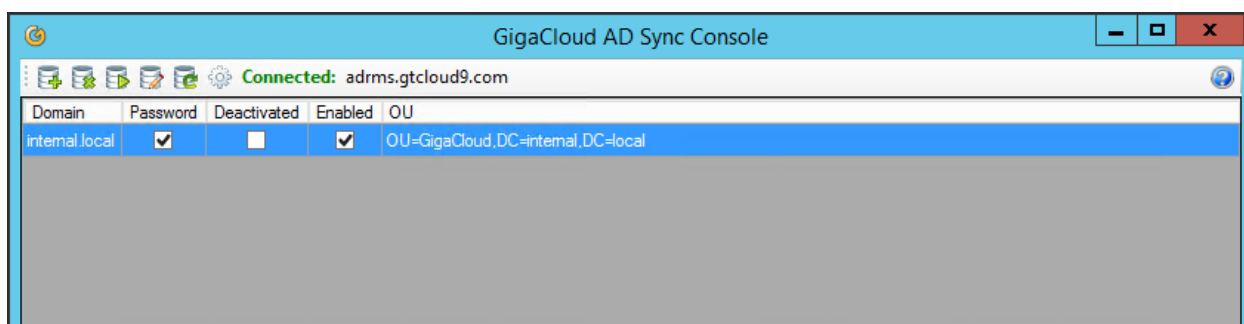
- h. If the prompt above is displayed, choose **Yes** to give the service account the permission to sync user passwords.



The dialog box titled "Enter Admin Credentials" has a blue header bar with a GigaCloud logo on the left and a close button (X) on the right. It contains three input fields: "Domain" with the text "company3.local", "Username" with the text "administrator@company3.local", and "Password" which is empty. At the bottom are "OK" and "Cancel" buttons.

Another dialog will be displayed asking for your domain service account credentials to add permission to access password hashes in the selected Active Directory domain. Enter your account credentials and select **OK**.

Note: You may need to close the console and reopen it for the permission changes to take effect.



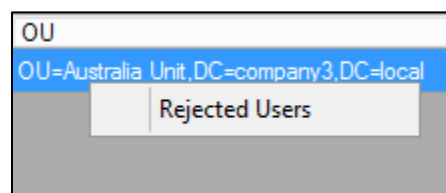
4. The OU you added will now be displayed in the upper panel in the console. In the lower panel messages will be displayed as the OU is configured in a local database and the first synchronization with GigaCloud is completed.



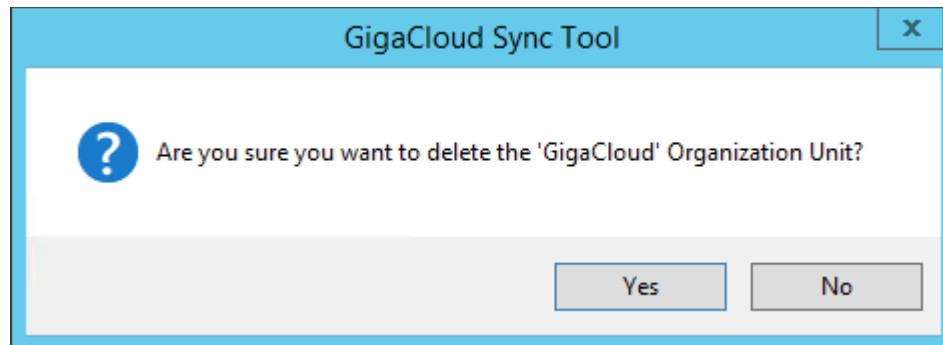
The Sync Agent, depending on connectivity and the resources on your server you have it installed on, will process approximately 1000 users a minute on first synchronization. On subsequent syncs, it will run much quicker because it will only update those objects which have changed since the last successful sync.

5. If you select the OU's row in the upper panel you can do the following:

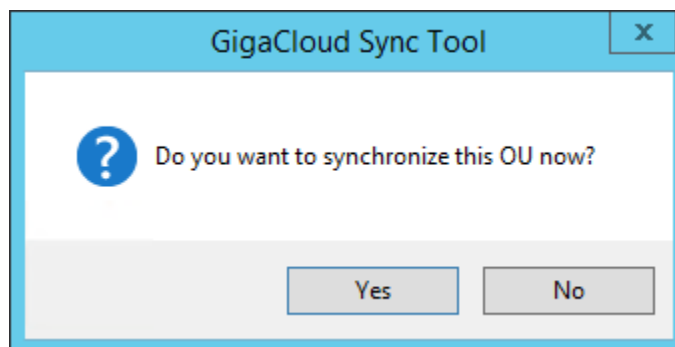
- a. If you right-click and choose **Rejected Users** you can view any users that were not synced. This is most often the case when a user belongs to another Tenant. Only one unique email address can reside in GigaCloud between all Tenants.



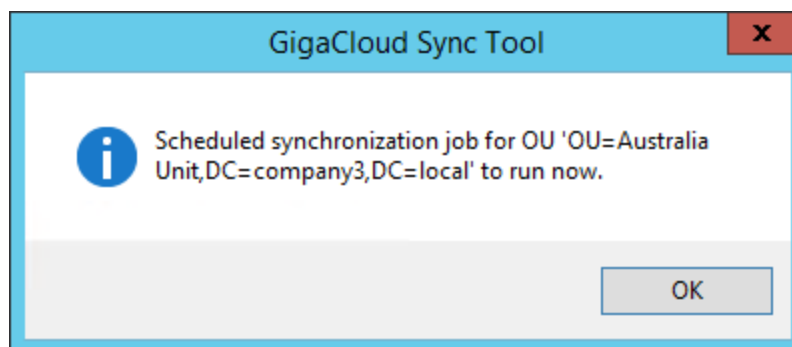
- b. If you need to delete the OU, select the  icon on the toolbar and confirm deletion. All objects in that OU will be immediately removed from GigaCloud. Although this is irreversible, you can readd the OU if needed. This only removes Users and Groups that are contained in your local OU from the Tenant container in GigaCloud.



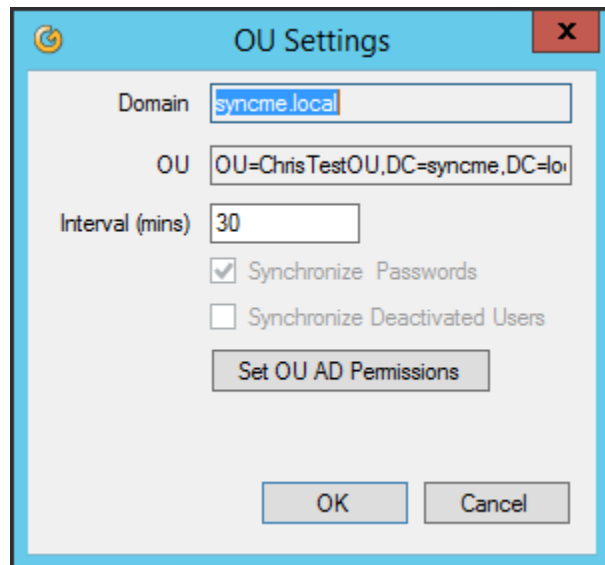
- c. Selecting this icon  puts a re-sync in the job queue for the selected OU. This is useful if you have made any local changes and need your GigaCloud Tenant updated.



Another confirmation dialog will be displayed. Select **OK** and sync jobs will be scheduled.



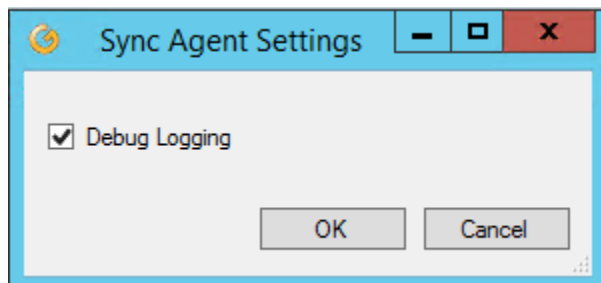
- d. The edit icon  will display the following dialog:



The image shows a Windows-style dialog box titled "OU Settings". It has a blue title bar with a yellow "G" icon on the left and a red "X" close button on the right. The dialog contains several input fields and checkboxes. The "Domain" field is set to "syncme.local". The "OU" field is set to "OU=ChrisTestOU,DC=syncme,DC=lo". The "Interval (mins)" field is set to "30". There are two checkboxes: "Synchronize Passwords" which is checked, and "Synchronize Deactivated Users" which is unchecked. Below these checkboxes is a button labeled "Set OU AD Permissions". At the bottom of the dialog are "OK" and "Cancel" buttons.

- **Domain-** displays the local forest domain name.
- **OU-** the CN of the selected OU.
- **Interval (mins)-** adjust the duration (or interval) in minutes that the OU is re-synced.
- **Synchronize Passwords (READ ONLY)-** indicates whether passwords are synced for this OU.
- **Synchronize Deactivated Users (READ ONLY)-** indicates whether locally deactivated users are synced.
- **Set OU AD Permissions-** This button will set the necessary Active Directory permission to allow syncing of the selected OU and parent OUs.

- e. The refresh icon  will reload the top OU panel.
- f. Selecting the settings icon  displays the **Settings** window where you can view and configure Sync Agent settings. The **Debug Logging** setting turns on and off additional Log Client logging which provides more detailed information for troubleshooting.



6. Continue to add other OUs (as needed) to sync with your Tenant account in GigaCloud.

 The Sync Agent only syncs Users and Groups from your domain controller and copies them to your Tenant Active Directory container in GigaCloud. The actual OUs are not synced. Each object synced is copied to your Tenant folder root and group membership is maintained for all imported users. If you delete a Group locally, on the next sync, that particular Group will be deleted in your Tenant folder as well as all user memberships to that group. Maintaining Group memberships in GigaCloud allows you to apply RMS Policy Templates and Rules at a group level, thereby simplifying your AD RMS policy management.

Group Naming

Since all objects in Active Directory must have a unique SamAccountName, to avoid a conflict with other Tenants Group names will automatically be appended with your Tenant name.

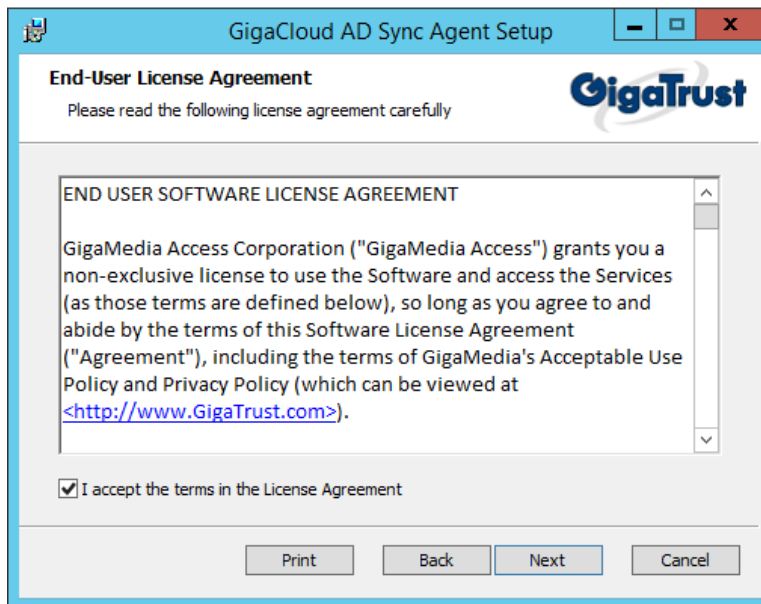
For example:

1. You have a group named “**Administration**” and it is synced with an OU from your Enterprise into GigaCloud.
2. The name “**Administration**” will be appended with “@[your domain name].[your domain extension]”
3. So if your forest domain is “**company2.com**”, when you search for the group in GTEP Administration the name will be displayed as “**Administration@company2.com**”.

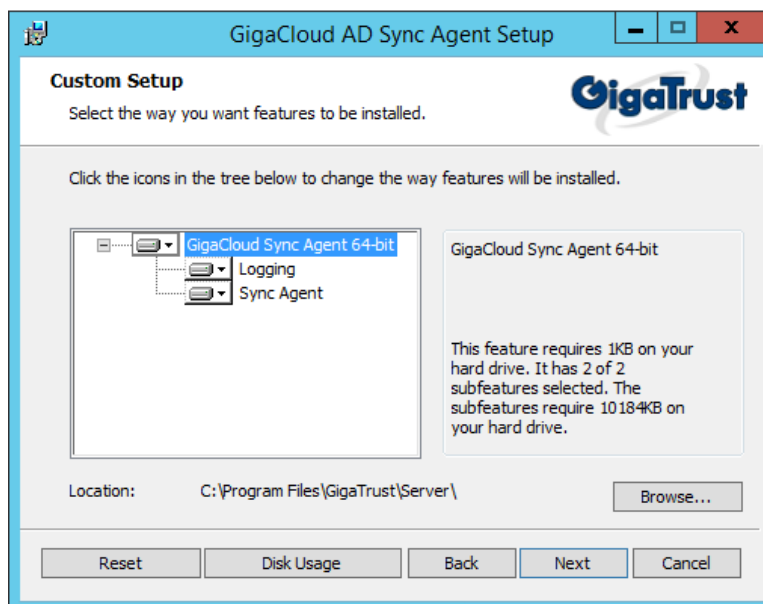
Upgrading

GigaCloud AD Sync Agent does not need to be uninstalled prior to upgrading. Follow the steps below to update your GigaCloud Sync Agent software:

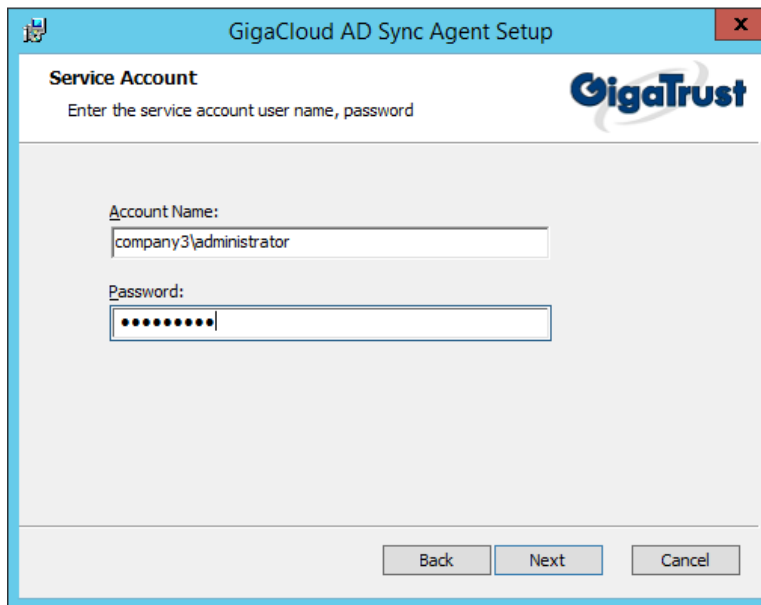
1. Click on the GigaCloud AD Sync Agent to start installation. Click **Next**.



2. Accept the End User Software License Agreement and select **Next**.

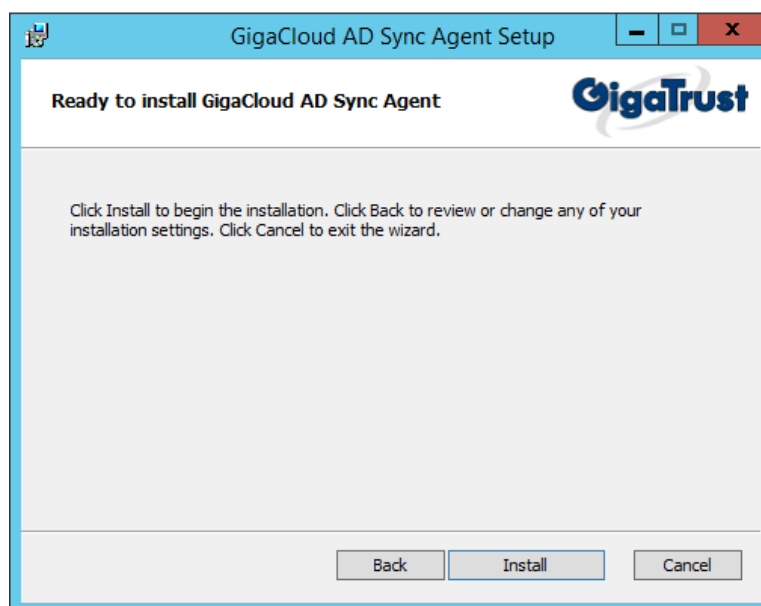


3. Leave both default features selected and click **Next**.

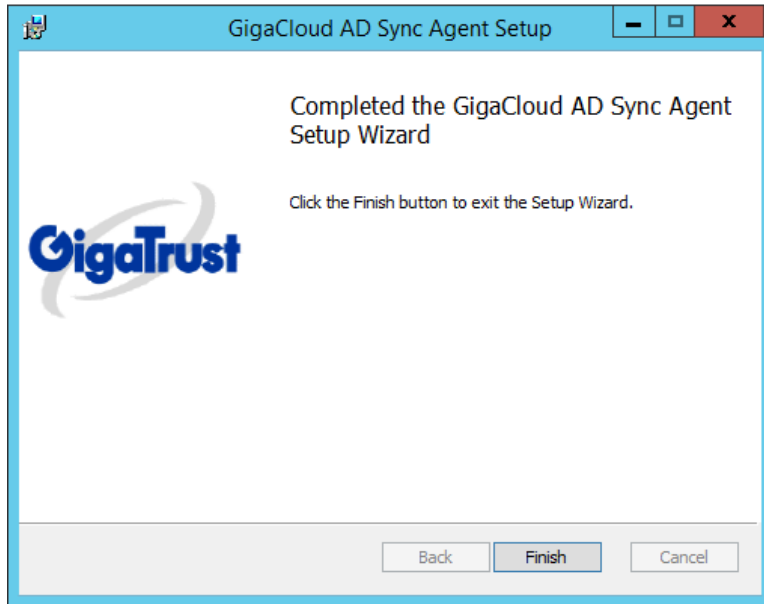


4. Enter the Domain User Account Name (this should be pre-populated from the back-up) and the Password and select Next.

This user should be a Domain Admin that is also a local Administrator and has local permissions to “logon as a service”.



5. Click **Install**.



6. Click **Finish**. Your software has been upgraded.

Troubleshooting

It is recommended that you use the GigaTrust Log Client for any troubleshooting. It will give you real time errors and the status of the GigaCloud Active Directory Sync Agent. This component can be optionally installed with the AD Sync Agent.

GigaTrust Log Client

All GTEP components send messages related to errors encountered and current status to the GigaTrust Log Server. The GTEP Log Client displays messages that are sent to the GigaTrust Log Server. You can install the GigaTrust Log Client component on its own server to isolate this component from the other GTEP components thereby allowing an administrator to delegate this process.

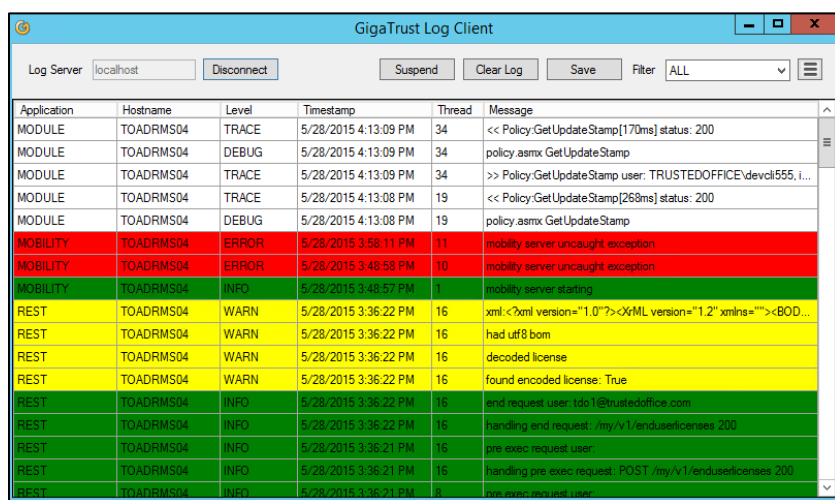
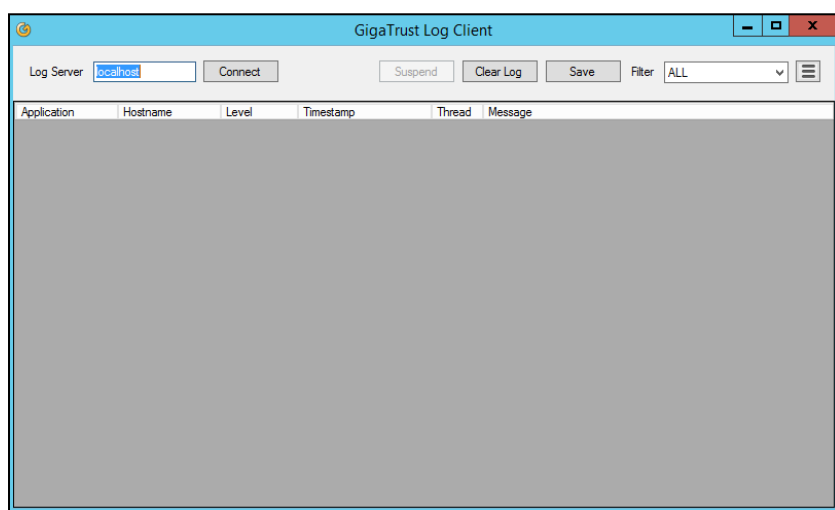
Access the GigaTrust Log Client via **All Programs > GigaTrust > GigaTrust Log Client**. Click **Connect** to view messages from the GigaTrust Log Server that is specified in the Log Server field. The default is *localhost*, but the GTEP Log Client can view messages from any accessible GigaTrust Log Server running in the network.

Click **Suspend** button to pause log messages from loading so you can view the details more closely. Click **Resume** when you are finished and the log files will continue loading from where it left off.

Click **Clear Log** to clear the messages from the display.

Click **Save** to write the currently displayed log messages to a file. A destination path and filename will be required for saving.

Export allows for saving log messages for a specific date/time range to a file. *From* and *To* dates, and time values, must be specified for the export. A destination path and filename will be required for saving.



The **Filter** drop-down list contains the different levels of messages that are displayed by the GigaTrust Log Client: *All*, *Debug*, *Info*, *Warn*, *Error*, and *Fatal*. Selecting a filter will display incoming messages received by the GigaTrust Log Server that are that level and above. (Display of existing messages is not

affected.) For example, if *Warn* is selected as the filter, only newly received messages with a status of *Warn*, *Error*, or *Fatal* will be displayed in the GTEP Log Client window.

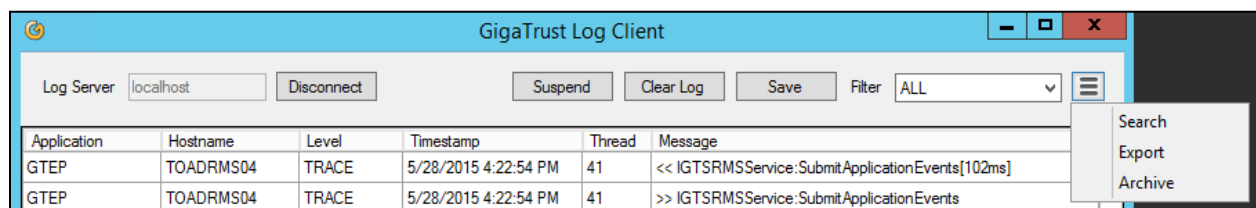
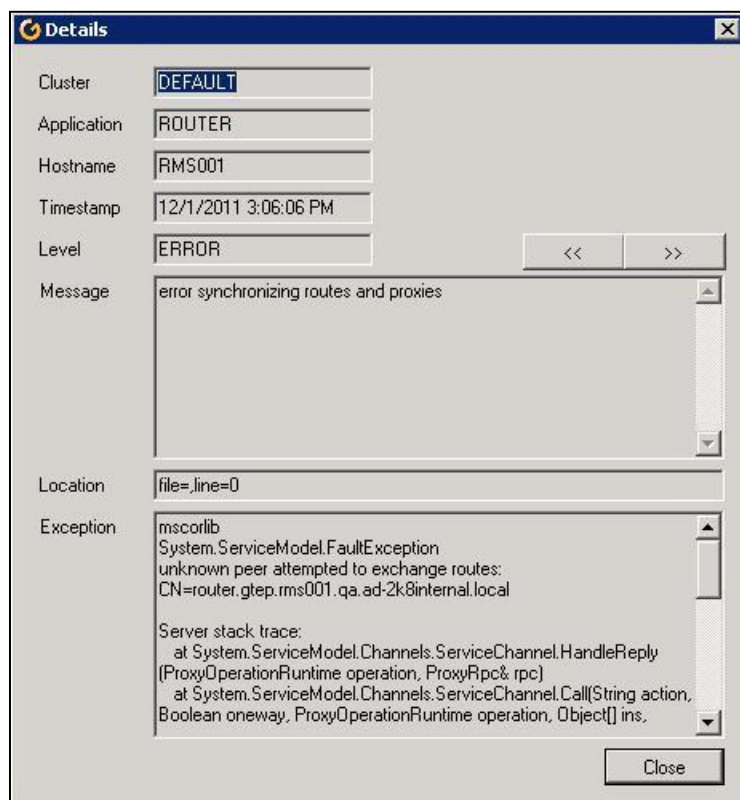
To view the details of any message, double-click its row in the GTEP Log Client window.

The top of the details window displays all field information that displays in the GTEP Log Client window. Previous (<<) and Next (>>) buttons are available to move between messages.

For Error level messages, the trapped error message is displayed in the Exception field at the bottom of the Details window. This information is critical to troubleshooting and resolving issues with the GTEP application.

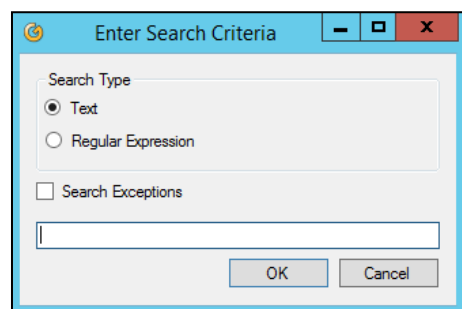


For additional features, click the menu icon at the upper right of the window.



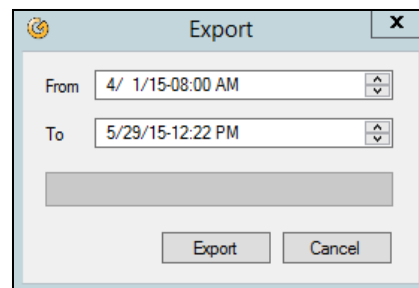
Search

Enter either text or a regular expression to search for matching results. Optionally, only search for Exceptions. Results are displayed with an option to **Save**.



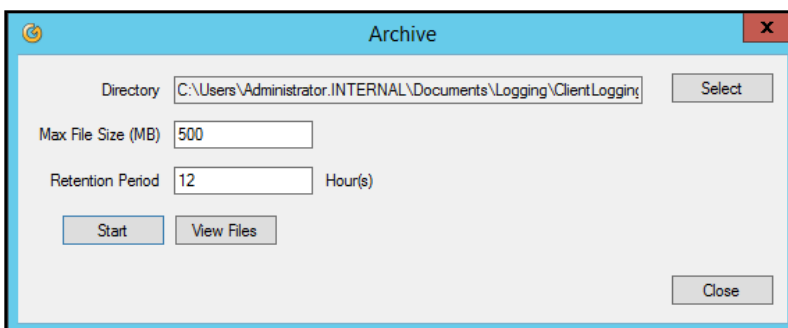
Export

Enter a date/time range for log messages to export. Specify the output filename and destination. The gray field shows progress as the export text file is created.



Archive

The Log Client display is limited to a rolling 1000 entries. As additional log entries are created, older entries will be discarded. When diagnostic efforts need more than the current 1000 entries, the Archive feature can be used. Archive will prompt for an external location to store the log data. This location is NOT limited to 1000 entries. The size of each log file is specified, in addition to the retention period. Enter a destination for the archive file. You can also optionally define the maximum log file size.



Click **Start** when you want to start collecting log messages and **Stop** to cease collecting.

A text file labeled GTLogClient<date><start time> is created in the specified destination folder.

Firewall Setting

To allow other GigaCloud Log servers to forward events to a dedicated Log Server port 10483 must be opened in the firewall.

Issues

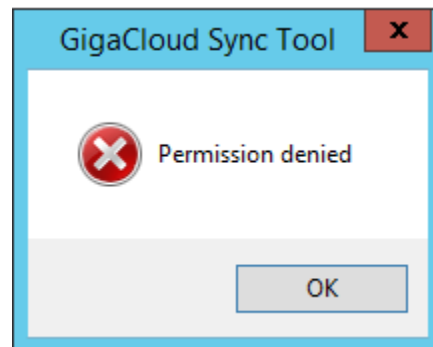
The following are some common issues and the resolution for each known problem you may encounter.

- **Issue:** On initial Activation you receive the “Permission denied” error.

Solution: This error can be caused by the following:

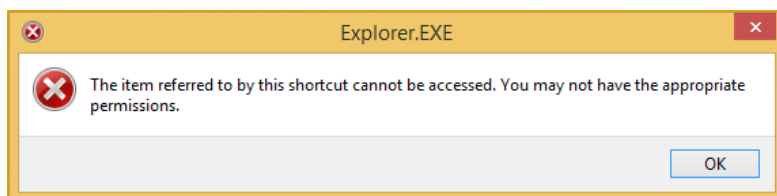
1. Invalid GigaCloud Email address. The email address provided must be the email used to create the GigaCloud Tenant.
2. The GigaCloud email address account has not yet been activated. Check your email and follow the instructions to activate your account.
3. Incorrect Password. Reenter and try again.

- **Issue:** A user or users will not sync to the GigaCloud.



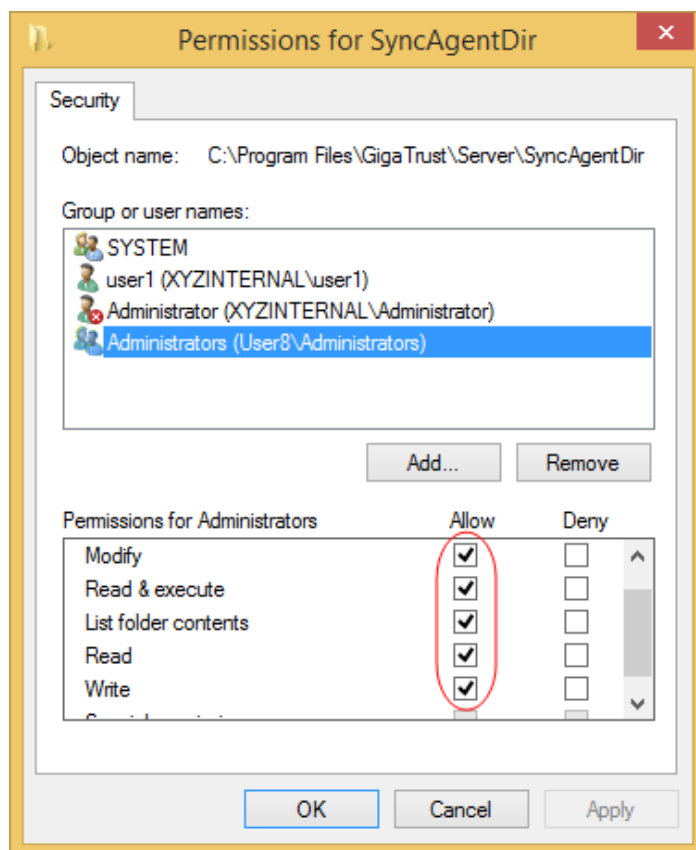
Solution: The user **MUST** have the **E-mail** attribute set in Active Directory. See [OU Requirements](#) for detailed instructions.

- **Issue:** When trying to access the application on a Windows desktop installation you receive the following error:



Solution: Use the following steps to give the appropriate directory permissions to the user affected: (Windows 7 shown; other platforms are similar)

1. "Go to the GigaTrust\Server\SyncAgentDir" folder where the application was installed. For example: "C:\Program Files\GigaTrust\GigaTrust\Server\SyncAgentDir".
2. Right click on the **SyncAgentDir** folder and select **Properties**. Select the **Security** tab and choose **Advanced**. When prompted, enter Domain Administrator credentials.
3. Select **Add** and on the resulting screen search for the current user and click **OK**.
4. Next add the required permissions for the user. The following should be checked:
 - a. Modify
 - b. Read & execute
 - c. List folder contents
 - d. Read
 - e. Write
5. Select **OK**.



Appendix

Active Directory Object Attributes Synchronized

The following tables list the Active Directory objects read from your Enterprise domain controller, and which attributes are synced in GigaCloud Active Directory. Some attributes are stored in the GigaCloud database. For instance, if Password synchronization is selected, passwords are synced and stored hashed in the GigaCloud database.

For your reference, the Sync Agent reads the following Active Directory object attributes from your Enterprise:

Object Type	Synchronized Object/Attribute
Enterprise	
	Forest Names
	Domain Names
	Forest SIDs
	Domain SIDs
	OU Distinguished Names
Group	
	groupType
	Member Of
	mail
	sAMAccountName
	objectSid
User	
	mail
	Member Of
	userAccountControl
	proxyAddresses
	givenName
	sn
	telephoneNumber